



Monitoring Virtual Visual Performance

Follow these steps to evaluate and interpret the virtual visual data provided by the **GCS software**:

1. Connect to the Drone:

- Ensure the drone is powered on and establish a connection with the ground control station software. This usually involves setting up a telemetry link via a radio transmitter/receiver, WiFi, or another communication method compatible with both the GCS and the drone.

2. Access the GCS Software:

- Launch the ground control station software on your computer or mobile device. Popular options include Mission Planner for ArduPilot, QGroundControl for PX4, and DJI Assistant for DJI drones.

3. View the Telemetry Data:

- Navigate to the **telemetry** or flight data section within the GCS software. This section provides real-time data from the drone, including altitude, speed, GPS position, battery voltage, and sensor readings.

4. Monitor Virtual Visual Performance:

- Look for visual indicators or graphical elements in the GCS software that represent the drone's position, orientation, and movement in a virtual environment. These may include:
 - **Map Display:** A feature that overlays the drone's position on a map or satellite imagery, allowing you to track the flight path and geographic location.
 - **HUD (Head-Up Display):** Provides a cockpit-style view showing altitude (pitch, roll, yaw), speed, and other flight parameters.
 - **Camera Feed:** Displays a live video feed from the drone's camera or FPV system, offering a real-time perspective from the drone.



5. Interpret the Visual Feedback:

- Assess the visual feedback from the GCS software to evaluate the drone's performance and behaviour. Look for **anomalies**, irregularities, or warning signs that could indicate operational issues, such as deviations from the flight path, unexpected movements, or abnormal sensor readings.

6. Troubleshoot and Adjust:

- If issues are identified with the drone's virtual visual performance, use the troubleshooting tools and features available in the GCS software to diagnose and resolve the problem. This might involve adjusting flight parameters, recalibrating sensors, or modifying the flight plan or mission parameters.

7. Monitor Continuously:

- Keep an ongoing watch on the drone's virtual visual performance throughout the flight operation. Regularly review telemetry data, visual indicators, and camera feed (if available) to ensure safe and reliable operation and respond promptly to any unexpected events or emergencies.